Corn Timeline

Corn: article

Domestication and Early Development

Circa 8000 BCE: humans used traditional modification methods like selective and cross-breeding to result in agriculture with desirable traits.
6,000-10,000 years ago: Native Americans living in what is now Mexico began domesticating teosinte
4,000 years ago: Northern Flint Corn (species grown by indigenous communities and tribes against the northern part of the U.S.)

Hybridization & Scientific Foundations

1800s: Two predominant races of corn of North America's eastern seaboard crossed (the late-maturing Virginia Gourdseed and the early-maturing Northeastern Flints) → crossed repeatedly, it ultimately created the Corn Belt dents (the most productive race of corn)
1835: newspaper articles started explaining how to cross varieties and gave directions for detasseling to produce hybrids.
1847: Robert Reid develops **Reid's Yellow Dent** through cross-pollination of flint and dent varieties.
1865: Gregor Mendel's discoveries about phenotypic variation
1866: Gregor Mendel identifies the basic process of genetics through the cross breeding of peas.
1878: Dr. W.J. Beal conducts hybrid corn experiments at Michigan State, leading to **Mule Corn**.
1982: FDA approves the first consumer GMO product
1986: Federal government establishes the Coordinated Framework for the Regulation of Biotechnology (policy describes how FDA, EPA, and USDA work together to regulate the safety of GMOs)
1887: New Bt trait = *Striacosta albicosta developed* (western bean cutworm)
1988-1989: New Bt trait = *Diabrotica* spp (spotted cucumber beetle)
1892 & 1893: George Morrow confirmed Beal's results (of mule corn) at the University of Illinois

Modern Breeding & Hybrid Expansion

1901: Discovery of **Bt trait** (insect resistance).
1908: G.H. Shull proposed that corn could be improved by selfing plants to develop inbred (the progeny from a self-fertilized plant is identical to the parent, and identical to one another) lines, making hybrids or crosses between the inbred lines and making that seed available for farmers.
1908 & 1909: Edward M. East worked with hybrids that yielded higher than 200 bu/acre
1917: D.F. Jones produces the **Burr-Leaning double-cross hybrid**.
1921: Burr-Leaning double cross hybrid was commercially produced
1922: First commercial hybrid corn (**Double-Cross 250**) sold by Funk's.

1927: discovery of X-ray induced mutations caused over 2,250 plant varieties to be developed
1934-36: U.S. Dust Bowl (hybrid seeds were more drought resistant)
1938: Bt proteins started being used in France
1939: Geneticist George Beadle demonstrated that teosinte grains could be heated and popped like popcorn
1940: Plant breeders learn to use radiation/chemicals to change an organism's DNA
1940s: white corn production dropped to 17% and was at 1% by 1970
1944: 2,4-D was developed, providing producers with their first opportunity to use selective weed control
1945: increased usage of inorganic nitrogen (N) fertilizer
1940s-70s: cateto corn varities collected across Brazil, Argentina, Uruguay, and Chile
Late 1940s: broadcast soil application of organochlorine insecticides (chemical control pests)
1949: Hybrid seeds reach **50–90% adoption** across U.S. corn production.
1950: about 90% of corn production area was planted with hybrid seeds
1950s: maize cultivation from Eastern Colorado into Nebraska

Biotechnology & Genetic Engineering

1953: Chemist Rosalind Franklin, and scientist James Watson and Francis Crick identified the structure of DNA.
Late 1950s: triazine herbicides were introduced
1958: Bt proteins started being used in the USA
1958: triazine herbicides were registered and changed weed management in maize
1959: control failures were noted and resistance spread with the concurrent corn rootworm range expansion (causes large yield reductions)
1960s: first nationally popular double cross hybrid (first generation hybrid between two single crosses) U.S.13 was used in breeding nurseries
(during the double cross era, national corn production averages increased by an average of 1 bu/acre each year)
1960s: production of single cross hybrid (first generation hybrid between two inbred lines) seeds became common
1964: DeKalb XL45, a first single cross hybrid, with a relative maturity of 110 days was first grown commercially (first early, popular hybrid released in the Northern U.S. Corn Belt)
1970s: maize cultivation in Indiana
1973: Biochemists Herbert Boyer and Stanley Cohen develop genetic engineering (GE) by inserting DNA from one bacteria into another.
1972: restriction enzymes were used to create the first recombinant DNA (to manipulate and isolate DNA segments of interest)
1973: Transgenic era (cost of pesticides went down, while seed prices escalated)
1973: Cohen et al. produced the first genetically modified organism, a bacteria
1980s: production of single cross hybrid seeds became the standard
(single cross hybrids were higher yielding and the annual national yield increased by an average of 1.71 bu/acre each year during the era)

1983: first genetically modified plant when Bevan et al. introduced a gene for antibiotic resistance to tobacco plants using the bacterium *Agrobacterium tumefaciens*
1990s: maize cultivation in the East Coast
1990s: first wave of GMO produce created through GE became available for consumers
1992: maize cultivation in Europe
1992: New Bt (insect control) trait = *Diatraea grandiosella developed*
1995: Environmental Protection Agency (EPA) in USA approved commercial production and distribution of Bt crops
1995: change in the teosinte branched1 (tb1) gene

1. tb1 gene controls apical dominance in maize plants by producing mRNAs that repress organ growth
    a. The mutation in the gene changed plant architecture to give domesticated corn one relatively strong stalk while teosinte had relatively weaker stalks.
        i. having one centralized stalk forced large concentrated amounts of resources into the stalk (including grain)

1996: Monsanto commercializes **Roundup Ready corn**; Mycogen & Ciba release first commercial **Bt corn hybrids**.
1996: glyphosate-resistant crops (GRCs) introduced (better control of weed spread)
1996: Mycogen seeds released the first commercial Bt-corn hybrids in partnership with Ciba Seeds (now part of Syngenta)
1996: First GM maize producing a Bt Cry protein was approved, killing the European corn borer and related species
1997: Corn with Bt trait was released commercially under the brand name YieldGard®
1997: Roundup Ready corn received FDA approval
1997: broad-spectrum, non-selective herbicide was released as Liberty Link corn by AgrEvo (now Bayer)
May 1998: Use of Starlink (Cry9C protein) in animal feed only was approved
1998: Roundup Ready corn received commercial release (provided the first stacked trait combination in corn//glyphosate tolerant corn)
2000: Target Induced Local Lesions In Genomes (TILLING which allowed for more targeted breeding approaches)
2000: Bt traits were 1% of total acreage
2002: Philippine government approved Bt corn
2003: The World Health Organization (WHO) and the Food and Agriculture Organization (FAO) of the United Nations develop international guidelines and standards to develop standards of GMO foods.
2003: Bt traits for rootworm control were released
2005: New Bt trait = Cry 3Bb (control of rootworms)
2006: armyworm resistance was first discovered in Puerto Rico
2007: South Africa researchers announced the production of transgenic maize resistant to maize streak virus (MSV)
2009: Government of Mexico created a regulatory pathway for GM maize
2009: Bt traits for rootworm resistance was documented (western corn rootworm?)

2010: USDA announced the deregulation of a new corn trait (Agrisure Viptera, industry's first non-crystalline protein for insect control) from Syngenta Seeds, Inc.
2010: genetically engineered crops were planted on 140 million hectares in 29 different countries
2010: Cuba developed a variety of genetically modified corn that is resistant to the palomilla moth
2011: Drought tolerant corn was developed by USDA (trait produces bacterial cold shock proteins (CSPs) that allow cellular machinery to function properly during periods of stress)
2013: 26 varieties of herbicide-resistant GM maize were authorized for import into the EU.
2013: Monsanto launches **DroughtGard®**, first commercial drought-tolerant transgenic corn. (MON 87460 trait is provided by the insertion of the cspB gene from the soil microbe Bacillus subtilis; it was approved by USDA in 2011 and by China in 2013)
2013: Drought tolerant corn was commercialized by USDA
2013: Bt traits were 17% of total acreage
2014: A few MSV-tolerant cultivars released in Africa (a private company named *Seedco* has released 5 MSV cultivars)
2014-2017: Researchers at Corteva Agriscience fused *zmm28* with a new promoter which consistently promoted corn yield by 3% to 5% in 48 varying commercial types of corn.
May 3, 2021: Penn State researchers discovered the Maize ufo1 gene responsible for creating the mutant corn line, affects cell differentiation, influences carbohydrates and hormone accumulation in the plant, and modulates expression patterns of essential genes involved in corn seed development
2021: Finding that the deletion of the WX1 locus produces a superior waxy corn
March 30, 2023: Nebraska researchers identify genes that help corn to adapt to new environments
May 28, 2024: ZmP LS6 in corn verified by scientists led by Iowa State University to regulate plant growth

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Soybean Timeline

Domestication & Early Use

11th century: domestication of soybean production in eastern half of North China
1712: Europeans were aware of soybeans through writing of German botanist
1740: Soybean seed may have been sent from China by missionaries and planted in France
1804: First use of the word **"soybean"**in U.S. literature, but may have been known as "Chinese vetches" in the colonies (1765)
1879: Rutgers Agricultural College in New Jersey published a soybean testing
1891: Japanese began to import soybean oil cake for use as fertilizer

Expansion & Commercialization

1908: Soybean shipments made to Europe and soybeans attracted worldwide attention
1920's: Soybean acreage expanded in great quantity to the U.S. corn belt
Before WWII: Soybeans were a successful crop as it solved the immediate need for soybean oil and meal
Post-WWII: Soybean production moved from the Southern U.S.
1950-1970s: U.S. dominated the soybean production by growing more than 75% of the world soybean crop
1970s: Discovery of protoplast transfection to study gene function in various plant species

Early Genetic & Disease Research

1970s: Worldwide shortage of protein led to the initiation of large-scale soybean production in several South American countries (especially Argentina and Brazil)
1979: Soybean Cyst Nematode (SCN) detected in Delaware
1986: *Glycine gracilis* discovered (a preliminary evolutionary product toward domesticated soybeans and G. soja as the progenitor of both G. gracilis and G. max)
1987: Biolistic plant transformation introduced by Klein, Wolf, Wu, & Sanford
1991: SCN (a disease that causes significant yield loss) discovered in New Castle County

Biotechnology & Genetic Engineering

1994: GTS 40-3-2 was approved commercially in the U.S.
1995: GTS 40-3-2 was approved commercially in Canada
1996: GTS 40-3-2 was approved commercially in Japan and Argentina
1997: GTS 40-3-2 was approved commercially in Uruguay
1998: First reported transgenic soybean plant using *Agrobacterium*-mediated DNA transfer (methods such as particle bombardment as well as the agrobacterium-tumefacien system and agrobacterium rhizogenes were used)
1998: GTS 40-3-2 was approved commercially in Brazil and Mexico
1999: GMO soybeans approved by the United Nations
2001: GTS 40-3-2 was approved commercially in South Africa
2003: The U.S. share of the world's soybean shrunk from 67% to 34% of world soybean production
2007: Romania imposed a ban upon GM soy entry into the EU

2010: Monsanto developed a glyphosate-resistant soybean that expresses Cry1Ac protein through crossing of MON87701 and MON89788
2010: DuPont Pioneer created a high oleic fatty acid soybean with levels of oleic acid greater than 80% to improve soybean oil
2011: herbicide-tolerant soybeans occupied 47% of the global biotech area, making it the principal biotech crop.
2014: U.S. contributed to 34% of world soybean production

Advanced Genetic Engineering & Gene Editing

2015: Bioceres obtained approval of HB4 soybeans in Argentina
2015: The first variety of glyphosate resistant generic soybeans developed at the University of Arkansas Division of Agriculture came to market
October 21, 2016: Drought-tolerant event, called HB4 or IND-ØØ41Ø-5 (herbicide tolerant, tolerant to stress and water deficit), developed by Argentine was presented and receives authorization from China
May 2019: HB4 soybeans approved in Brazil
August 2019: HB4 soybeans approved in the U.S.
2019: HB4 soybeans approved in Paraguay
December 17, 2020: CRISPR/Cas9 system used to induce knockouts into GmGOLS1a and GmGOLS1B genes to reducing RFO (Raffinose family oligosaccharides) levels by 35.4% in double mutant soybeans, improving soybean nutrition quality
2021: HB4 soybeans approved in Canada
2021: Cpf1 (Cas12a) systems established in soybean for genetic engineering
2021: GmMs1 KO events in soybeans were created (showing male sterility phenotype)
October 19, 2021: Gene editing of GmJAGGED1 (which controls the number of seeds per pod), increasing China's soybean yield by 8%-10%
February 20, 2022: WI12Rhg1 interacts with DELLAs found with GE to for SCN-resistant mechanisms
April 29, 2022: Chinese Ministry of Agriculture announced the approval of HB4
June 2022: *Ochrobactrum haywardense*-mediated delivery method used to efficiently target soybean genes

Table of soybean traits altered using genome editing technology:
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